

Marketing Funnel & ROI Analytics Dashboard

Marketing Analytics / Business Analytics Project Case Study

Project Type

Business Analytics / Marketing Analytics

Tools & Technologies

Excel | Python | Power BI | DAX

Key Areas

Marketing Analytics | Data Visualization | KPI Analysis | Business Insights

Project Case Study

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MARKETING FUNNEL & ROI ANALYTICS DASHBOARD

Analyzing Marketing Performance Across Channels, Campaigns & Time

1. Project Title & Overview

1.1. Project Overview

This project analyzes marketing funnel performance across multiple digital marketing channels and campaigns using Excel, Python and Power BI. The analysis tracks the complete marketing journey from impressions and clicks to leads, conversions and revenue, while evaluating campaign efficiency through key performance indicators such as CTR, CPC, CPL, Conversion Rate, CAC and ROI.

An interactive Power BI dashboard was developed to enable analysis of marketing performance by channel, campaign and month. The project aims to transform raw marketing data into actionable insights that can support marketing budget allocation, campaign optimization and data-driven business decisions.

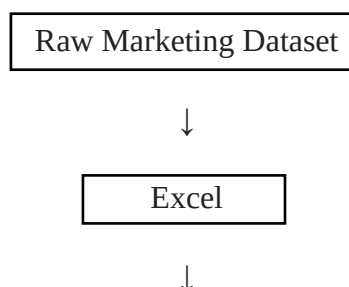
- i. **Project Type:** Business Analytics / Marketing Analytics
- ii. **Domain:** Digital Marketing & Business Analytics
- iii. **Project Level:** End-to-End Analytics Project

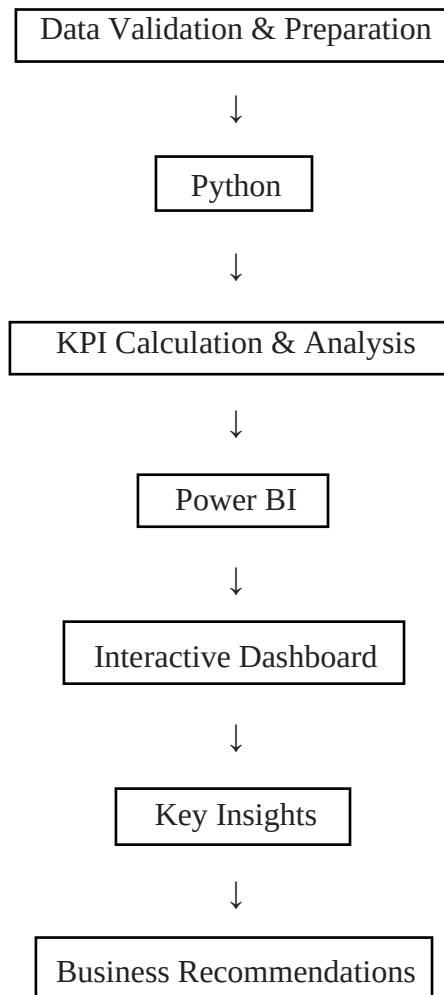
1.2. Tools Used

Tools & Technologies

- **Microsoft Excel** — Dataset management and initial validation
- **Python (Pandas, NumPy)** — Data cleaning, validation and analysis
- **Power BI** — Interactive dashboard and visualization
- **DAX** — KPI measures and calculations

1.3. Project Architecture





2. Business Problem

2.1. What is the Business Problem?

Digital marketing campaigns generate large amounts of data across different channels, campaigns and time periods. However, having data alone does not provide a clear understanding of which marketing activities are generating the strongest business results. The key challenge is to evaluate the complete marketing funnel—from impressions and clicks to leads, conversions and revenue—and determine how efficiently marketing expenditure is being converted into business outcomes. Without a consolidated view of metrics such as CTR, CPC, CPL, Conversion Rate, CAC and ROI, it can be difficult to identify high-performing channels, inefficient campaigns and opportunities for improving marketing investment. Therefore, this project focuses on analyzing marketing performance across multiple channels and campaigns and developing an interactive dashboard that provides a consolidated view of marketing spend, funnel performance, revenue, customer acquisition cost and return on investment.

2.2. Key Business Questions

- 1) Which marketing channels generate the highest revenue?
- 2) Which channels provide the highest ROI?
- 3) Which channels have the lowest customer acquisition cost (CAC)?
- 4) Which campaigns generate the highest number of leads and conversions?
- 5) Which campaigns provide the strongest ROI?
- 6) How does marketing revenue change over time?
- 7) How efficiently do prospects move through the marketing funnel?
- 8) Where are the major opportunities for improving marketing performance?
- 9) How can marketing spending be better evaluated using revenue, conversions, CAC and ROI?

2.3. Primary Business Objective

To analyze marketing funnel performance and identify patterns in channel, campaign and time-based performance that can support data-driven marketing decisions related to budget allocation, campaign optimization and customer acquisition efficiency.

3. Project Objectives

3.1. Objective

The primary objective of this project is to evaluate marketing funnel and campaign performance using data-driven analysis and develop an interactive dashboard that enables stakeholders to monitor marketing efficiency and identify opportunities for optimization.

3.2. Specific Objectives

1. Analyze Marketing Funnel Performance:

Analyze the progression of users through the marketing funnel from impressions → clicks → leads → conversions and evaluate the efficiency of each stage.

2. Evaluate Channel Performance:

Compare marketing channels based on spend, impressions, clicks, leads, conversions, revenue, CAC and ROI to understand differences in performance and efficiency.

3. Analyze Campaign Performance:

Evaluate individual campaigns using key metrics such as revenue, conversions, ROI and customer acquisition cost to identify variations in campaign performance.

4. Measure Marketing Efficiency:

Calculate and analyze CTR, CPC, CPL, Conversion Rate, CAC and ROI to measure how effectively marketing expenditure is being converted into measurable business outcomes.

5. Analyze Revenue Trends:

Examine monthly marketing revenue and conversion trends to identify periods of stronger or weaker performance.

6. Build an Interactive Power BI Dashboard:

Develop an interactive Power BI dashboard that allows users to analyze marketing performance using Date, Channel and Campaign filters.

7. Generate Actionable Business Insights:

Translate analytical findings into business insights and recommendations related to marketing budget allocation, campaign optimization, customer acquisition efficiency and funnel performance.

3.3. Analytical Scope

The analysis covers marketing performance across five channels:

- Google Ads
- Meta Ads
- Instagram
- Facebook
- Email Marketing

The analysis evaluates ten marketing campaigns across the period January 2025 to August 2026 using 1,000 marketing records.

3.4. Expected Outcome

The project aims to provide a consolidated view of marketing performance and enable data-driven evaluation of channel efficiency, campaign performance, customer acquisition cost, revenue generation and return on investment.

4. Dataset Description

4.1. Dataset Overview

The analysis was performed using a structured marketing dataset containing 1,000 records of marketing campaign activity. The dataset covers the period from January 2025 to August 2026 and includes performance data across multiple marketing channels and campaigns.

Each record contains information related to marketing reach, engagement, expenditure, lead generation, conversions and revenue. These fields were used to evaluate the complete marketing funnel and calculate key marketing performance indicators.

4.2. Dataset Summary

Attribute	Description
Number of Records	1,000
Number of Columns	15
Time Period	January 2025 – August 2026
Marketing Channels	5
Marketing Campaigns	10
Primary Currency	INR (₹)
Data Type	Structured marketing performance data

4.3. Marketing Channels

The dataset contains performance information for the following five marketing channels:

- Google Ads
- Meta Ads
- Instagram
- Facebook
- Email Marketing

4.4. Marketing Campaigns

The dataset contains ten campaign categories:

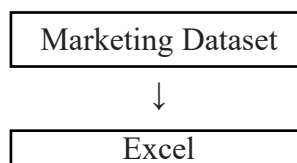
- i. Summer Sale
- ii. Festive Offers
- iii. New Product Launch
- iv. Brand Awareness
- v. Lead Generation
- vi. Monsoon Campaign
- vii. Diwali Promotion
- viii. Year-End Sale
- ix. Retargeting Campaign
- x. Customer Acquisition

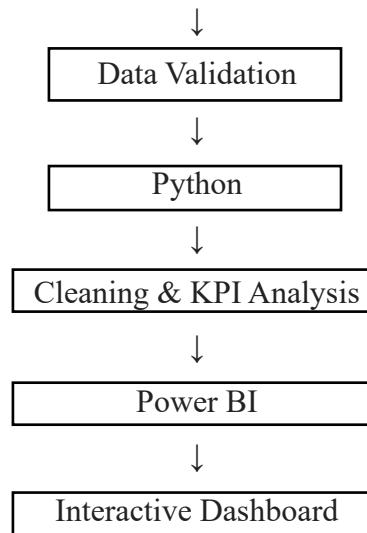
4.5. Dataset Fields

Field	Description
Date	Date associated with the marketing activity
Campaign	Marketing campaign name
Channel	Marketing channel used
Impressions	Number of times the marketing content was displayed
Clicks	Number of clicks generated
Ad Spend (INR)	Marketing expenditure
Leads	Number of leads generated
Conversions	Number of successful conversions
Revenue (INR)	Revenue attributed to the marketing activity
CTR (%)	Click-through rate
CPC (INR)	Cost per click
CPL (INR)	Cost per lead
Conversion Rate (%)	Percentage of leads converted
CAC (INR)	Customer acquisition cost
ROI (%)	Return on marketing investment

4.6. Dataset Flow

The dataset was initially reviewed and validated in Excel before being imported into Python for data cleaning, validation and analytical calculations. The prepared data and calculated metrics were then used in Power BI to develop the interactive marketing analytics dashboard.





4.7. Data Quality Checks

The dataset was validated using Python to check:

- Dataset dimensions
- Column names and data types
- Missing values
- Duplicate records
- Negative values in key numerical fields
- Date format consistency
- KPI calculation consistency

The dataset contained **1,000 records with no intended missing or duplicate records**, and the calculated KPI values were validated against the corresponding dataset metrics.

5. Tools & Technologies

5.1. The project uses a combination of spreadsheet tools, Python libraries and business intelligence technologies to clean, analyze and visualize marketing performance data.

Tool / Technology	Purpose
Microsoft Excel	Used for initial dataset inspection, data organization, validation and spreadsheet-based analysis.
Python	Used for data cleaning, validation, KPI calculation and exploratory data analysis.
Google Colab	Used as the cloud-based environment for executing Python code and analyzing the dataset.
Pandas	Used for data manipulation, filtering, grouping, aggregation and preparation of marketing data.

Tool / Technology	Purpose
NumPy	Used for numerical calculations and handling mathematical operations during analysis.
Power BI	Used to create the interactive marketing performance dashboard and visualize business insights.
DAX	Used in Power BI to create calculated measures for marketing KPIs such as total spend, leads, conversions, revenue, CAC and ROI.

5.2. Technology Workflow

Excel → Python / Pandas / NumPy → Data Validation & KPI Analysis → Power BI / DAX → Interactive Dashboard

Role of Each Technology:

- Excel: Organized and inspected the original marketing dataset.
- Python: Performed data preparation, validation and analytical calculations.
- Pandas and NumPy: Supported data manipulation and numerical analysis.
- Power BI: Converted analytical results into interactive charts, KPI cards, slicers and tables.
- DAX: Created dynamic measures that respond to dashboard filters and selections.

6. KPI Calculations & Formulas

To evaluate marketing funnel performance and campaign efficiency, several key performance indicators (KPIs) were calculated from the marketing dataset. These metrics help measure engagement, acquisition efficiency, conversion performance and return on marketing investment.

6.1. Click Through Rate (CTR)

CTR measures the percentage of impressions that resulted in clicks.

Formula:

$$\text{CTR (\%)} = (\text{Clicks} \div \text{Impressions}) \times 100$$

Purpose:

Used to evaluate how effectively marketing campaigns generate user engagement from impressions.

6.2. Cost Per Click (CPC)

CPC measures the average amount spent to generate one click.

Formula:

$$\text{CPC} = \text{Ad Spend} \div \text{Clicks}$$

Purpose:

Used to evaluate the cost efficiency of generating traffic through marketing campaigns.

6.3. Cost Per Lead (CPL)

CPL measures the average marketing expenditure required to generate one lead.

Formula:

$$\text{CPL} = \text{Ad Spend} \div \text{Leads}$$

Purpose:

Used to compare lead-generation efficiency across channels and campaigns.

6.4. Conversion Rate

Conversion Rate measures the percentage of generated leads that resulted in conversions.

Formula:

$$\text{Conversion Rate (\%)} = (\text{Conversions} \div \text{Leads}) \times 100$$

Purpose:

Used to evaluate how effectively leads are converted into customers or successful outcomes.

6.5. Customer Acquisition Cost (CAC)

CAC measures the average marketing expenditure required to generate one conversion.

Formula:

$$\text{CAC} = \text{Ad Spend} \div \text{Conversions}$$

Purpose:

Used to evaluate customer acquisition efficiency and compare acquisition costs across marketing channels and campaigns.

6.6. Return on Investment (ROI)

ROI measures the return generated from marketing expenditure after accounting for the amount spent.

Formula:

$$\text{ROI (\%)} = ((\text{Revenue} - \text{Ad Spend}) \div \text{Ad Spend}) \times 100$$

Purpose:

Used to evaluate the financial efficiency of marketing campaigns and channels.

6.7. Funnel Conversion Metrics

The marketing funnel was evaluated across four major stages:

Impressions → Clicks → Leads → Conversions

The funnel helps identify how effectively prospects move from initial exposure to the final conversion stage.

Additional funnel indicators include:

- **Click Rate:** $\text{Clicks} \div \text{Impressions}$
- **Lead Rate:** $\text{Leads} \div \text{Clicks}$
- **Lead-to-Conversion Rate:** $\text{Conversions} \div \text{Leads}$
- **Overall Funnel Conversion:** $\text{Conversions} \div \text{Impressions}$

These metrics provide a detailed understanding of where potential customers are progressing or dropping off within the marketing funnel.

KPI Analysis Objective:

The calculated KPIs were used to compare marketing performance across **channels, campaigns and time periods**. The results were subsequently incorporated into the Power BI dashboard to enable interactive analysis and support data-driven marketing evaluation.

7. Data Cleaning & Preparation

Before performing detailed analysis and visualization, the marketing dataset was inspected and prepared using Python. The objective of this stage was to ensure that the data was correctly structured, consistent and suitable for KPI calculations and dashboard development.

7.1. Data Loading

The Excel dataset was imported into Python using the Pandas library. The **Marketing Data** worksheet containing the 1,000 marketing records was loaded for analysis.

7.2. Dataset Structure Validation

The dataset structure was examined to verify:

- Number of rows and columns
- Column names
- Data types
- Date field structure
- Numerical fields
- Categorical fields

The dataset contained **1,000 records and 15 columns**.

7.3. Data Conversion

The **Date** column was converted into an appropriate date format to enable time-based analysis.

This allowed the dataset to be analyzed by:

- Day
- Month

- Year
- Monthly marketing performance

7.4. Missing Value Check

All columns were checked for missing or null values.

The purpose of this validation was to ensure that missing data would not affect KPI calculations or dashboard visualizations.

7.5. Duplicate Record Check

The dataset was checked for duplicate records to ensure that the same marketing activity was not unintentionally counted more than once.

7.6. Numerical Data Validation

Key numerical fields were examined for invalid or negative values.

The following fields were validated:

- Impressions
- Clicks
- Ad Spend
- Leads
- Conversions
- Revenue

This helped ensure that the numerical values were logically suitable for marketing performance analysis.

7.7. KPI Validation

The existing KPI values in the dataset were independently recalculated using Python.

The following metrics were validated:

- CTR
- CPC
- CPL
- Conversion Rate
- CAC
- ROI

The calculated values were compared with the corresponding KPI fields to verify consistency.

7.8. Categorical Data Validation

Marketing channels and campaign names were reviewed to ensure consistency in category values.

The analysis covered:

5 Marketing Channels:

- Google Ads
- Meta Ads
- Instagram
- Facebook
- Email Marketing

10 Marketing Campaigns:

- Summer Sale
- Festive Offers
- New Product Launch
- Brand Awareness
- Lead Generation
- Monsoon Campaign
- Diwali Promotion
- Year-End Sale
- Retargeting Campaign
- Customer Acquisition

7.9. Data Preparation Outcome

After validation, the dataset was considered suitable for further exploratory analysis and Power BI visualization.

The prepared dataset was then used to analyze marketing performance across **channels, campaigns and time periods**, and to develop the interactive Power BI dashboard.

8. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed using Python to understand the distribution, patterns and performance of marketing activities before developing the Power BI dashboard.

The analysis focused on marketing performance across **channels, campaigns and time periods**.

8.1. Overall Marketing Performance

The overall dataset was analyzed to understand the total scale of marketing activity.

Key metrics examined included:

- Total Impressions
- Total Clicks
- Total Ad Spend
- Total Leads
- Total Conversions
- Total Revenue
- Overall CTR
- Overall Conversion Rate
- Overall CAC
- Overall ROI

This provided a baseline view of overall marketing performance before analyzing individual channels and campaigns.

8.2. Channel-Level Analysis

Marketing performance was compared across the five channels:

- Google Ads
- Meta Ads
- Instagram
- Facebook
- Email Marketing

The following metrics were analyzed for each channel:

- Ad Spend
- Impressions
- Clicks
- Leads
- Conversions
- Revenue
- CAC
- ROI

The channel-level analysis helped identify differences in marketing efficiency, revenue generation and customer acquisition cost.

8.3. Campaign-Level Analysis

The ten marketing campaigns were analyzed to compare their performance.

Key metrics included:

- Leads generated
- Conversions

- Revenue
- CAC
- ROI

This analysis helped identify campaigns with stronger conversion and financial performance and provided a basis for understanding differences between campaign strategies.

8.4. Time-Based Analysis

Marketing performance was analyzed over time to identify changes in revenue and conversion activity.

Monthly analysis was performed using:

- Monthly Revenue
- Monthly Conversions
- Monthly Leads
- Monthly Ad Spend
- Monthly ROI
- Monthly CAC

This helped identify periods of higher and lower marketing performance and supported further investigation of monthly trends.

8.5. Funnel Analysis

The complete marketing funnel was examined using four major stages:

Impressions → Clicks → Leads → Conversions

The analysis evaluated the movement of prospects through each stage and calculated funnel conversion metrics.

This helped identify the overall efficiency of the marketing journey from initial exposure to final conversion.

8.6. Key EDA Observations

The exploratory analysis revealed differences in performance across marketing channels, campaigns and time periods.

The analysis showed that:

- Marketing channels differed considerably in revenue generation and acquisition efficiency.
- ROI and CAC varied across channels and campaigns.
- Lead volume did not necessarily correspond directly to the highest financial efficiency.
- Marketing performance changed across different months.
- The funnel showed substantial drop-off between impressions, clicks, leads and final conversions.

These observations were used to guide the development of the Power BI dashboard and the subsequent business insights.

9. Power BI Dashboard Development

After completing data preparation and exploratory analysis, the prepared marketing dataset was imported into Power BI to create an interactive dashboard for monitoring marketing funnel and campaign performance.

The dashboard was designed to provide a consolidated view of marketing expenditure, funnel activity, revenue generation, customer acquisition efficiency and return on investment.

9.1. Data Import

The prepared marketing dataset was imported into Power BI.

The data model was reviewed to ensure that:

- Date fields were recognized correctly.
- Numerical fields were available for aggregation.
- Marketing channels and campaigns could be used as filters.
- Revenue and Ad Spend were treated as currency values.
- KPI measures could be calculated dynamically.

9.2. DAX Measures

DAX measures were created to calculate important marketing KPIs dynamically.

The primary measures included:

- Total Ad Spend
- Total Leads
- Total Conversions
- Total Revenue
- CTR
- CPC
- CPL
- Conversion Rate
- CAC
- ROI

These measures allow the dashboard calculations to automatically respond to selected dates, channels and campaigns.

9.3. Dashboard Layout

A one-page interactive dashboard was developed with the following major components:

1. Dashboard Header

A professional header was created to display the project title and provide a consistent visual identity.

2. Interactive Filters

Slicers were added for:

- Date
- Channel
- Campaign

These filters allow users to analyze specific periods, marketing channels or campaigns.

3. KPI Cards

KPI cards were used to display key overall performance indicators:

- Total Ad Spend
- Total Leads
- Total Conversions
- Total Revenue
- ROI

4. Marketing Funnel

A funnel visualization was created to represent the customer journey:

Impressions → Clicks → Leads → Conversions

This provides a visual representation of the movement of prospects through different stages of the marketing funnel.

5. Revenue by Channel

A chart was created to compare revenue generated by the five marketing channels.

6. ROI by Channel

A separate visualization was used to compare ROI across marketing channels and evaluate differences in financial efficiency.

7. Monthly Revenue Trend

A time-series visualization was created to monitor changes in marketing revenue across months.

8. Campaign Performance Table

A detailed table was created to compare campaign-level performance using metrics such as:

- Campaign
- Leads
- Conversions
- Revenue
- CAC
- ROI

9.4. Dashboard Interactivity

The dashboard was designed to support interactive analysis.

Users can select a specific:

Date → Channel → Campaign

and the KPI cards and visualizations dynamically update based on the selected filters.

This allows stakeholders to move from an overall marketing performance view to more detailed channel- and campaign-level analysis.

9.5. Dashboard Design Principles

The dashboard was designed with the following principles:

- Clear KPI hierarchy
- Simple and consistent layout
- Minimal visual clutter
- Consistent typography
- Clear chart titles
- Appropriate use of whitespace
- Interactive filtering
- Easy comparison of marketing performance

9.6. Dashboard Outcome

The final Power BI dashboard provides an interactive view of marketing funnel performance and enables users to analyze marketing expenditure, leads, conversions, revenue, CAC and ROI across different channels, campaigns and time periods.

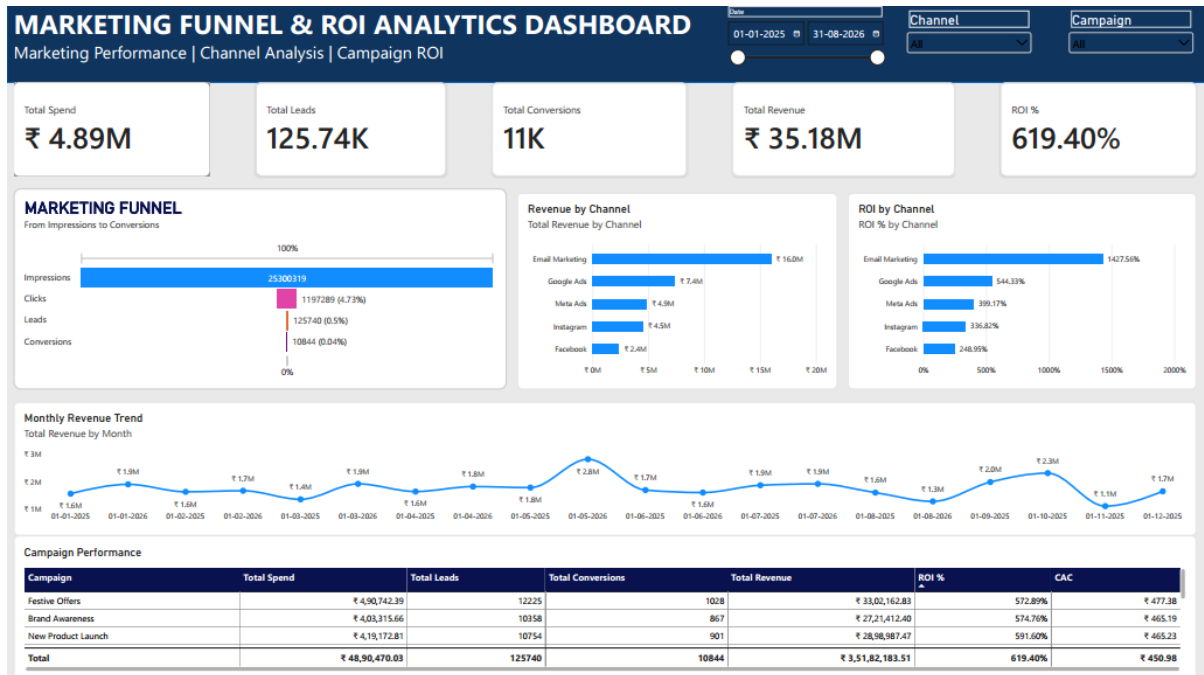


Figure 1: Marketing Funnel & ROI Analytics Dashboard

10. Python Analysis & Results

Python was used to perform exploratory analysis and validate the marketing dataset before developing the Power BI dashboard. The analysis was carried out using **Pandas** and **NumPy** in Google Colab.

The Python workflow focused on understanding the dataset structure, validating the marketing KPIs and comparing performance across channels and campaigns.

10.1. Dataset Inspection

The Excel dataset was loaded into Python using Pandas.

Initial inspection was performed using:

- Dataset dimensions
- Column names
- Data types
- Missing-value checks
- Duplicate checks
- Descriptive statistics

The dataset contained **1,000 records** and **15 columns**.

10.2. KPI Recalculation and Validation

Marketing KPIs were independently calculated in Python using the underlying marketing data.

The following metrics were recalculated:

- CTR
- CPC
- CPL
- Conversion Rate
- CAC
- ROI

The recalculated KPI values were compared with the corresponding values in the dataset to validate the consistency of the analytical data.

10.3. Channel Level Analysis

Python was used to group and aggregate the dataset by marketing channel. Channel-level analysis included:

- Total Ad Spend
- Total Leads
- Total Conversions
- Total Revenue
- ROI
- CAC

This analysis enabled comparison of marketing performance across Google Ads, Meta Ads, Instagram, Facebook and Email Marketing.

10.4. Campaign Level Analysis

The dataset was also grouped by campaign to evaluate campaign performance.

The analysis compared:

- Lead generation
- Conversion volume
- Revenue
- ROI
- CAC

This helped identify differences in performance among the ten marketing campaigns.

10.5. Time Based Analysis

Date-based aggregation was performed to examine marketing performance over time.

Monthly metrics were analyzed to identify changes in:

- Revenue
- Leads
- Conversions
- Ad Spend

- ROI
- CAC

This analysis provided the foundation for the monthly performance visualization used in Power BI.

10.6. Python Analysis Workflow

The overall Python analysis followed the workflow:

Load Dataset → Inspect Data → Validate Data → Recalculate KPIs → Analyze Channels → Analyze Campaigns → Analyze Monthly Trends → Prepare Data for Visualization

10.7. Analysis Output

The Python analysis provided validated and aggregated information that was subsequently used to build the interactive Power BI dashboard.

The analysis established a quantitative foundation for evaluating marketing funnel performance, channel efficiency, campaign performance and financial returns.

11. Analysis Results & Key Insights

The analysis of the marketing dataset provided insights into overall funnel performance, channel-level revenue and ROI, campaign performance and changes in marketing revenue over time.

11.1. Overall Marketing Performance

The dashboard provides the following overall performance metrics for the analyzed period:

KPI	Result
Total Ad Spend	₹4.89M
Total Leads	125.74K
Total Conversions	10,844
Total Revenue	₹35.18M
Overall ROI	619.40%

The marketing funnel consisted of **25,300,319 impressions, 1,197,289 clicks, 125,740 leads and 10,844 conversions.**

This indicates that the analysis covered the complete customer journey from initial marketing exposure through engagement, lead generation and final conversion.

11.2. Channel Level Results

The analysis showed differences in revenue generation and ROI across the five marketing channels.

Channel	Revenue	ROI
Email Marketing	₹16.0M	1,427.56%
Google Ads	₹7.4M	544.33%
Meta Ads	₹4.9M	399.17%
Instagram	₹4.5M	336.82%
Facebook	₹2.4M	248.95%

Email Marketing generated the highest revenue among the analyzed channels at approximately **₹16.0M** and recorded an ROI of **1,427.56%**. Google Ads generated approximately **₹7.4M** in revenue with an ROI of **544.33%**.

The ROI comparison also showed substantial variation between channels, ranging from **248.95% for Facebook to 1,427.56% for Email Marketing**.

11.3. Campaign Level Results

Campaign performance was evaluated using Ad Spend, Leads, Conversions, Revenue, ROI and CAC.

The dashboard provides the following campaign-level results for the campaigns displayed in the report:

Campaign	Spend	Leads	Conversions	Revenue	ROI	CAC
Festive Offers	₹4,90,742	12,225	1,028	₹33,02,163	572.89%	₹477.38
Brand Awareness	₹4,03,316	10,358	867	₹27,21,412	574.76%	₹465.19
New Product Launch	₹4,19,173	10,754	901	₹28,98,987	591.60%	₹465.23
Monsoon Campaign	₹6,12,904	15,714	1,363	₹43,43,799	608.72%	₹449.67

The displayed campaign records show differences in lead volume, conversion volume, revenue, ROI and acquisition cost. Monsoon Campaign, for example, generated **15,714 leads and 1,363 conversions** in the displayed campaign analysis.

11.4. Funnel Analysis

The marketing funnel was analyzed across four stages:

Impressions → Clicks → Leads → Conversions

The dashboard shows:

- **25,300,319 Impressions**
- **1,197,289 Clicks**
- **125,740 Leads**
- **10,844 Conversions**

The corresponding displayed rates include approximately **4.73% from impressions to clicks**, **0.50% from clicks to leads**, and **0.04% from impressions to conversions**.

The funnel analysis helps identify the scale of activity at each stage and highlights the reduction in volume as prospects move toward the final conversion stage.

11.5. Monthly Revenue Analysis

Monthly revenue was visualized using a time-series chart covering the project period.

The monthly trend shows that revenue varied across different months, with several periods showing higher revenue levels than others. The dashboard provides monthly revenue values ranging from approximately **₹1.1M to ₹2.8M** in the displayed trend.

This time-based analysis can be used to investigate changes in marketing performance and identify periods requiring further analysis.

11.6. Key Insights

Based on the analysis, the following insights were identified:

Insight 1 — Significant variation in channel ROI

ROI varied substantially across marketing channels, indicating differences in financial efficiency between channels.

Insight 2 — Email Marketing generated substantial revenue

Email Marketing generated approximately **₹16.0M** in revenue and recorded **1,427.56% ROI** in the analyzed dataset.

Insight 3 — Google Ads contributed significant revenue

Google Ads generated approximately **₹7.4M** in revenue and recorded **544.33% ROI**, making it an important contributor to the analyzed marketing revenue.

Insight 4 — Campaign performance differed across KPIs

Campaigns showed differences in leads, conversions, revenue, ROI and CAC. This demonstrates why campaign evaluation should consider multiple performance indicators rather than relying on a single metric.

Insight 5 — Funnel volume decreases substantially toward conversion

The large difference between impressions and final conversions demonstrates significant movement reduction across the marketing funnel. This creates opportunities for further analysis of engagement, lead generation and conversion stages.

11.7. Business Interpretation

The analysis demonstrates that marketing performance should be evaluated using a combination of **revenue, conversions, CAC and ROI**, rather than focusing only on impressions, clicks or leads. The differences observed across channels and campaigns provide a basis for investigating marketing efficiency, optimizing campaign strategies and improving the conversion journey.

12. Business Recommendations

Based on the analysis of marketing funnel performance, channel-level results, campaign performance and ROI, the following recommendations can be considered to improve marketing efficiency and support data-driven decision-making.

12.1. Evaluate Budget Allocation by ROI and CAC

Marketing investment should be evaluated using a combination of **ROI, CAC, conversions and revenue** rather than relying on a single metric.

Channels and campaigns demonstrating stronger financial efficiency can be studied further to understand the factors contributing to their performance.

12.2. Optimize High-CAC Channels

Channels with relatively higher customer acquisition costs should be investigated to identify opportunities for improving efficiency.

Possible areas of optimization include:

- Audience targeting
- Campaign creatives
- Ad messaging
- Landing-page experience
- Lead qualification
- Conversion follow-up

The objective should be to reduce acquisition costs while maintaining or improving conversion volume.

12.3. Investigate High-ROI Channel Strategies

Email Marketing recorded the highest ROI in the analyzed dataset.

The strategies contributing to this performance can be examined further, including:

- Audience segmentation
- Campaign messaging
- Customer targeting

- Follow-up communication
- Repeat engagement

Insights from these activities can be used as a reference when evaluating other marketing initiatives.

12.4. Maintain Focus on Conversion Efficiency

High lead volume does not necessarily translate into proportionally high conversions.

Marketing teams should therefore monitor the complete funnel:

Impressions → Clicks → Leads → Conversions

Improving lead quality, landing-page experience and follow-up processes can help improve movement between funnel stages.

12.5. Analyze Campaign-Level Performance

Campaigns should be evaluated using multiple KPIs, including:

- Leads
- Conversions
- Revenue
- CAC
- ROI

This provides a more complete understanding of campaign performance than evaluating campaigns based only on lead volume or engagement.

12.6. Investigate Monthly Performance Variations

The monthly revenue trend shows variations in marketing revenue across the analyzed period.

Periods with unusually high or low performance should be investigated further to identify possible contributing factors such as:

- Campaign activity
- Seasonal demand
- Marketing expenditure
- Channel mix
- Promotional activity
- Conversion performance

12.7. Improve Lower-Funnel Performance

The funnel analysis shows a substantial reduction in volume from impressions to final conversions.

Further optimization can focus on the lower stages of the funnel through:

- Landing-page optimization
- Lead qualification
- Retargeting
- Follow-up communication
- Conversion-focused messaging
- Customer journey improvements

12.8. Establish Continuous Marketing Performance Monitoring

The Power BI dashboard can be used as a recurring performance-monitoring tool.

Marketing teams can periodically track:

Spend → Leads → Conversions → Revenue → CAC → ROI

This enables performance changes to be identified earlier and provides a consistent framework for evaluating marketing activities.

Recommendation Summary

The analysis indicates that marketing decisions should be based on a combination of **financial performance, acquisition efficiency, funnel movement and campaign-level results**.

A structured monitoring approach using Power BI can help stakeholders continuously evaluate marketing performance and identify areas requiring further investigation and optimization.

13. Project Conclusion

This project successfully developed an end-to-end **Marketing Funnel & ROI Analytics Dashboard** using Excel, Python and Power BI.

The analysis evaluated marketing performance across multiple channels and campaigns by examining the complete funnel from **impressions and clicks to leads, conversions and revenue**. Key performance indicators including **CTR, CPC, CPL, Conversion Rate, CAC and ROI** were used to measure marketing engagement, acquisition efficiency and financial performance.

Python was used for data validation, KPI verification and exploratory analysis, while Power BI and DAX were used to transform the analyzed data into an interactive dashboard. The dashboard enables users to monitor marketing performance by **date, channel and campaign**.

The project demonstrated that marketing data can be transformed into meaningful business insights by combining data preparation, quantitative analysis and interactive visualization. The resulting dashboard provides a structured approach for evaluating marketing performance and identifying areas for further investigation and optimization.

Overall, the project strengthened practical skills in **Marketing Analytics, Business Analytics, Python, Excel, Power BI, DAX, KPI analysis and data-driven decision-making**.

14. Limitations & Future Scope

14.1. Project Limitations

Although the project provides a structured analysis of marketing funnel performance, several limitations should be considered when interpreting the results.

1. Dataset Scope

The analysis is based on **1,000 marketing records covering January 2025 to August 2026**. Therefore, the findings represent the performance contained within this dataset and should not automatically be generalized to other businesses, markets or time periods.

2. Limited Marketing Variables

The dataset primarily contains campaign performance metrics such as impressions, clicks, spend, leads, conversions and revenue. Additional variables such as customer demographics, audience segments, geographic information and detailed creative-level data are not included.

3. Attribution Limitations

The analysis uses the revenue and conversion values provided in the dataset. A detailed multi-touch attribution model was not implemented to determine how multiple marketing interactions contributed to an individual conversion.

4. External Factors

The analysis does not explicitly account for external factors such as competitor activity, market conditions, economic changes, seasonal demand or changes in customer behavior.

5. Causality

The analysis identifies patterns and performance differences in the dataset but does not establish that a particular marketing channel or campaign directly caused the observed results.

14.2. Future Scope

The project can be extended in several ways to provide deeper marketing and business insights.

1. Customer Segmentation

Customer-level data could be incorporated to analyze performance across demographic, geographic or behavioral segments.

2. Multi-Touch Attribution

A multi-touch attribution model could be implemented to understand the contribution of different marketing touchpoints throughout the customer journey.

3. Customer Lifetime Value

Customer Lifetime Value (CLV) could be incorporated alongside CAC to evaluate the longer-term financial value of acquired customers.

4. Predictive Analytics

Machine learning models could be introduced to forecast:

- Leads
- Conversions
- Revenue
- Customer acquisition costs
- Campaign performance

5. Marketing Budget Optimization

Optimization techniques could be used to simulate different budget allocation scenarios and evaluate their potential impact on revenue, conversions and ROI.

6. Real-Time Dashboard Integration

The Power BI dashboard could be connected to live marketing platforms or regularly refreshed data sources to enable more frequent performance monitoring.

7. Automated Reporting

Automated reporting could be introduced to distribute periodic marketing performance summaries to stakeholders.

Future Development

The project can therefore be expanded from a **descriptive marketing analytics dashboard** into a more advanced analytics solution incorporating **customer segmentation, attribution, predictive analytics, forecasting and marketing budget optimization**.

15. Project Workflow / End-to-End Methodology

The project followed a structured end-to-end analytics workflow, beginning with the marketing dataset and progressing through data validation, Python analysis, KPI evaluation, Power BI visualization and business interpretation.

15.1. End-to-End Workflow

Step 1 — Marketing Dataset

The structured marketing dataset containing campaign performance records was used as the primary data source.



Step 2 — Excel Data Inspection

The dataset was initially inspected and organized in Excel to understand its structure, fields and marketing metrics.



Step 3 — Data Validation & Preparation

The data was validated for structure, data types, missing values, duplicate records, numerical consistency and date formatting.



Step 4 — Python Analysis

Python, Pandas and NumPy were used to perform data inspection, KPI validation, aggregation and exploratory analysis.



Step 5 — KPI Calculation

Marketing performance was evaluated using:

CTR | CPC | CPL | Conversion Rate | CAC | ROI



Step 6 — Channel & Campaign Analysis

Marketing performance was compared across channels, campaigns and time periods using revenue, conversions, CAC and ROI.



Step 7 — Power BI Development

The prepared data was imported into Power BI and DAX measures were created for dynamic KPI calculations.



Step 8 — Interactive Dashboard

An interactive one-page dashboard was developed with KPI cards, marketing funnel, channel analysis, ROI analysis, monthly revenue trend, campaign performance table and slicers.



Step 9 — Insights & Interpretation

The dashboard and analytical results were used to identify performance patterns across channels, campaigns and time periods.



Step 10 — Business Recommendations

The identified insights were translated into recommendations related to marketing efficiency, funnel optimization, campaign evaluation and performance monitoring.

15.2. Methodology Summary

Raw Data → Excel → Data Validation → Python → KPI Analysis → Channel/Campaign Analysis → Power BI & DAX → Interactive Dashboard → Insights → Business Recommendations.